Clean Accuracy Explains Most of the Robustness Gap: A Controlled Comparison of ResNet-18 and VGG-16 under Adversarial Training on CIFAR-100

1st Marlen Oryntay
National Physics and Mathematics School
Almaty, Kazakhstan
oryntaymarlen@gmail.com

2nd Anwar Ghani
School of Computing and Artificial Intelligence
Nazarbayev University
Astana, Kazakhstan
anwar.ghani@nu.edu.kz

***Abstract*—Residual networks are often reported to be more robust than plain sequential networks after adversarial training, but such comparisons rarely separate robustness from clean accuracy. We train ResNet-18 and VGG-16 on CIFAR-100 under an identical recipe, with and without a mixed clean/adversarial training objective, and evaluate them with FGSM, PGD-7, PGD-20 and AutoAttack at ℓ_∞ budgets up to $\epsilon = 8/255$, averaging over several random seeds. Without defence, both networks reach close to 0% accuracy under PGD at $\epsilon = 4/255$. Adversarial training at $\epsilon = 4/255$ raises PGD-20 accuracy at that budget to 28.17% (ResNet-18) and 21.84% (VGG-16), at a clean-accuracy cost of 5.15 and 3.94 points, but transfers poorly to $\epsilon = 8/255$ (AutoAttack: 5.77% and 4.13%). ResNet-18 is ahead in absolute robust accuracy at every budget. However, when we decompose robust accuracy into clean accuracy times the fraction of correctly classified inputs that survive the attack, about 70% of the gap at $\epsilon \leq 4/255$ and all of it at $\epsilon = 8/255$ is explained by ResNet-18's higher clean accuracy. We also find that checkpoint selection on a small robust validation set adds substantial variance to the reported trade-off, and that the "failure cases are better learned" phenomenon recently reported for PGD-based adversarial training does not appear under our weaker objective, although adversarial training moves both networks towards it.**

***Index Terms*—Adversarial machine learning, adversarial training, robustness, accuracy-robustness trade-off, ResNet-18, VGG-16, CIFAR-100.**

I. Introduction

Convolutional Neural Networks (CNNs) reach high accuracy on image classification, yet their predictions can be changed by small, carefully crafted perturbations of the input [1], [2]. This is not only a laboratory curiosity: small stickers on a stop sign can mislead a traffic-sign classifier in the physical world [3], and similar attacks are a concern wherever an attacker can influence the input of a deployed model, for example in autonomous driving, medical imaging or automated inspection [4].

Many early defences were later broken by adaptive attacks, often because they only obfuscated gradients [4], [5]. Adversarial training (AT), i.e., training on perturbed inputs, remains the most reliable empirical defence [6], [7]. AT has two well-known costs. First, it lowers accuracy on clean data, the accuracy-robustness trade-off [8]–[12]. Second, its outcome depends on the network architecture [13], [14].

These two effects are entangled whenever architectures are compared. A prediction can only remain correct under attack if it was correct on the clean input, so a model that classifies more clean inputs correctly also has more inputs that can survive an attack. Benchmarks usually rank models by absolute robust accuracy [13], [15], and the most robust network is typically also the most accurate one. It is then unclear whether the architecture makes the model harder to attack, or simply classifies more inputs correctly to begin with. The first would justify choosing an architecture for robustness; the second only means that a stronger classifier stays stronger.

We study this question in a controlled setting. We choose ResNet-18 [16] and VGG-16 [17] as two classical representatives of residual and plain sequential CNNs, and CIFAR-100 as a dataset on which clean accuracy is far from saturated, so that the two networks differ clearly in clean accuracy. Both networks are trained with exactly the same optimiser, schedule, data augmentation and mixed clean/adversarial AT objective, over three random seeds, and are evaluated with single-step, multi-step and ensemble attacks at three perturbation budgets. Fig. 1 gives an overview of the pipeline. Our contributions are:

- A multi-seed benchmark of a residual and a sequential CNN on CIFAR-100, with and without mixed-loss AT, under FGSM, PGD-7, PGD-20 and AutoAttack [18], with sanity checks against gradient masking.
- A simple decomposition of robust accuracy into clean accuracy and conditional robust accuracy, showing that most of ResNet-18's robustness advantage is inherited from its higher clean accuracy.
- Two methodological observations: best-checkpoint selection on a small robust validation set makes the trade-off noisy, and the perception-consistency pattern of [12] does not appear under a weaker AT objective.

Section II reviews related work, Section III presents the methodology and algorithms, Sections IV and V give the setup and results, and Section VI concludes.

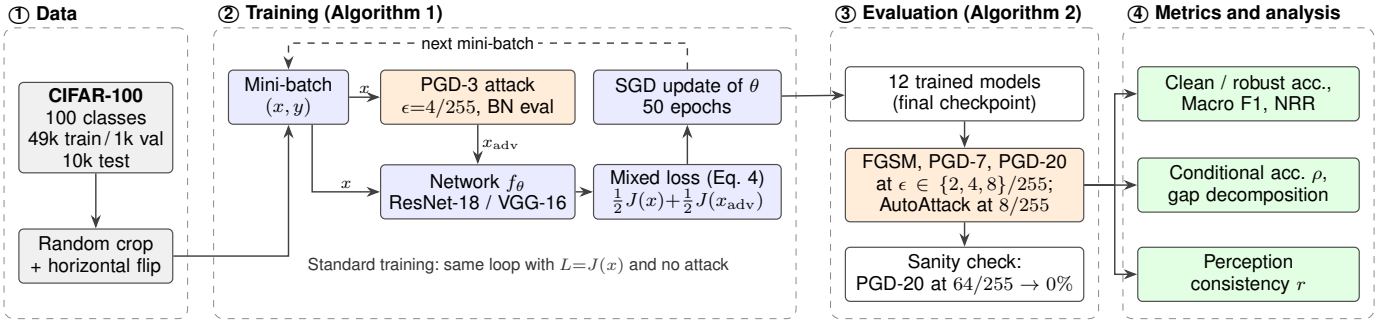


Fig. 1. Overall architecture of the study. ResNet-18 and VGG-16 are trained on CIFAR-100 with standard cross-entropy or with mixed-loss adversarial training on PGD-3 examples generated on the fly (Algorithm 1). The 12 trained models are attacked with FGSM, PGD and AutoAttack, and the results are summarised by accuracy, NRR, conditional robust accuracy ρ with the gap decomposition, and perception consistency (Algorithm 2).

II. Related Work

A. Adversarial Attacks and Robustness Evaluation

Szegedy et al. first showed that DNNs misclassify imperceptibly perturbed images [1]. Goodfellow et al. explained this effect by the locally linear behaviour of DNNs and proposed the single-step FGSM attack [2]; Madry et al. generalised it into the multi-step PGD attack, which approximates the worst case inside an ℓ_p ball [6]. Because many early defences appeared robust only against weak or poorly tuned attacks [4], [5], evaluation practice has moved towards stronger and standardised protocols. AutoAttack [18] combines several parameter-free attacks to avoid overestimating robustness, and RobustBench [15] tracks models evaluated in this way. Recent AT papers typically report clean accuracy, PGD-20 accuracy and AutoAttack accuracy averaged over several runs [12]; we follow the same protocol.

B. Adversarial Training and Its Variants

AT is formulated as a min-max problem in which the inner maximisation crafts adversarial examples and the outer minimisation fits the model to them [6]; Bai et al. survey the resulting family of methods [7]. Wong et al. show that FGSM training with random initialisation can approach PGD-based AT at a fraction of the cost, but may suffer from catastrophic overfitting [19]. Rice et al. identify robust overfitting, where robust test accuracy degrades late in training, and recommend early stopping on robust validation accuracy [20]. Adversarial weight perturbation flattens the weight loss landscape and improves robust generalisation [21], ReBAT reinterprets robust overfitting as an imbalance in the game between trainer and attacker [22], and ADR replaces one-hot targets with rectified soft labels obtained by self-distillation [23]. Our objective (Eq. (4)) is deliberately simple, a mix of clean and adversarial cross-entropy in the spirit of [2], so that differences between architectures are not hidden by a tuned recipe.

C. The Accuracy-Robustness Trade-off

Tsipras et al. argue that robustness and accuracy can be fundamentally at odds [8], and TRADES makes the trade-off explicit by splitting robust error into natural and boundary error [9]. The drop in clean accuracy under AT has commonly been attributed to an overly complex decision boundary caused by insufficient learning of hard adversarial examples [24], which motivated sample-adaptive methods such as limiting the excessive margin that AT builds along adversarial directions [10]. CURE selectively updates parts of the network to conserve clean knowledge and introduced the Natural-Robustness Ratio (NRR) to summarise the trade-off [11]. Wang and Liu [12] measure perception consistency as the mean squared error (MSE) between the logits of a clean input and its perturbed version, separately for successfully defended and failed samples. For PGD-AT with ResNet-18 on CIFAR-10 they find that failed samples have a smaller logit change than defended ones, i.e., failure cases are learned over-sufficiently, and they propose Robust Perception Adversarial Training (RPAT) to make logits change smoothly along the perturbation. We reuse their measurement to ask whether the same pattern holds for a weaker AT objective and for a plain sequential network.

D. Architecture and Robustness

Huang et al. study how depth and width should be allocated across residual stages for adversarially robust networks [13], and Peng et al. distil architectural design principles for robust CNNs from a large-scale study [14]. From a theoretical angle, architecture affects the Lipschitz constant and sensitivity of the model [25]. These works focus on modern or tuned architectures and compare models mainly by absolute robust accuracy. We instead compare two classical networks under one fixed recipe and explicitly separate the contribution of clean accuracy.

III. Methodology

A. Threat Model and Adversarial Training

Let f_θ be a classifier and $J(\theta, x, y)$ the cross-entropy loss. Under the white-box threat model the adversary knows the architecture and weights and searches for a perturbation δ with $\|\delta\|_\infty \leq \epsilon$ that maximises the loss. Adversarial training solves [6]

$$\min_\theta \ \mathbb{E}_{(x,y)}\Big[\max_{\|\delta\|_\infty \leq \epsilon} J(\theta, x+\delta, y)\Big]. \tag{1}$$

Pixel values lie in $[0, 1]$; input normalisation is the first layer of each model, so all perturbation budgets are expressed in pixel units.

B. Attacks

FGSM takes one signed-gradient step of size ϵ [2]:

$$x_{\text{adv}} = \text{clip}_{[0,1]}\big(x + \epsilon \cdot \text{sign}(\nabla_x J(\theta, x, y))\big). \tag{2}$$

PGD-K starts from a uniformly random point in the ϵ-ball $\mathcal{B}_\epsilon(x)$ and iterates [6]

$$x^{t+1} = \Pi_{\mathcal{B}_\epsilon(x)\cap[0,1]^d}\big(x^t + \alpha \cdot \text{sign}(\nabla_x J(\theta, x^t, y))\big). \tag{3}$$

We use PGD-7 with $\alpha = \epsilon/4$ and PGD-20 with $\alpha = 2/255$. AutoAttack (standard version: APGD-CE, APGD-T, FAB-T, Square) [18] is run at $\epsilon = 8/255$ on the first 1,000 test images.

C. Network Architectures

ResNet-18 [16] (11.22M parameters) stacks residual blocks with identity shortcuts; we use a 3×3 stem without initial max-pooling for 32×32 inputs. VGG-16 [17] (15.30M parameters) is a plain stack of 3×3 convolutions with Batch Normalisation and no shortcuts, followed by global average pooling and a classifier with two hidden layers of 512 units (Batch Normalisation, ReLU, dropout 0.5). The networks therefore differ in more than skip connections, which we take into account in Section V-D.

D. Mixed-Loss Adversarial Training

We minimise an equally weighted sum of clean and adversarial loss:

$$L = \tfrac{1}{2}J(\theta, x, y) + \tfrac{1}{2}J(\theta, x_{\text{adv}}, y), \tag{4}$$

where x_{adv} is generated for every mini-batch by PGD with 3 steps, $\epsilon = 4/255$ and $\alpha = 1/255$, with Batch Normalisation in inference mode while crafting the attack. The training budget is deliberately smaller than the largest evaluation budget, so that we can observe how robustness transfers to a budget not seen in training. Algorithm 1 lists the training steps; standard training uses the same loop with $L = J(\theta, x, y)$ and no attack.

E. Evaluation Protocol and Metrics

Every trained model is evaluated on the CIFAR-100 test set with the steps of Algorithm 2. We report (i) clean accuracy A_{clean}; (ii) robust accuracy A_{rob}, the top-1 accuracy on adversarial test images; (iii) the Macro F1-score over the 100 classes; (iv) the Natural-Robustness Ratio [11]

$$\text{NRR} = \frac{2\,A_{\text{clean}}\,A_{\text{rob}}}{A_{\text{clean}} + A_{\text{rob}}}, \tag{5}$$

the harmonic mean of the two accuracies; (v) the conditional robust accuracy

$$\rho = \Pr\big(f(x_{\text{adv}}) = y \,\big|\, f(x) = y\big), \qquad A_{\text{rob}} = A_{\text{clean}}\cdot\rho, \tag{6}$$

i.e., the fraction of correctly classified test images that remain correct under attack;[1] and (vi) perception consistency [12]: the mean logit MSE between each clean image and its PGD-20 version, computed separately for defended and failed samples among those classified correctly on clean data. We report the ratio $r = \text{MSE}_{\text{failed}}/\text{MSE}_{\text{defended}}$ (only when at least 30 samples are defended); $r < 1$ corresponds to the "failure cases are better learned" pattern of [12].

By Eq. (6), the gap in robust accuracy between the adversarially trained networks can be split exactly into a part due to clean accuracy and a part due to conditional robustness:

$$\Delta A_{\text{rob}} = \underbrace{\Delta A_{\text{clean}} \cdot \rho^{\text{VGG}}}_{\text{clean accuracy}} + \underbrace{A_{\text{clean}}^{\text{ResNet}} \cdot \Delta\rho}_{\text{conditional robustness}}. \tag{7}$$

[1]The identity holds when inputs misclassified on clean data remain misclassified under attack, which is the case for nearly all samples in our runs.

Algorithm 1 Mixed-loss adversarial training (one seed)

Input: training set $\mathcal{D}$, validation set $\mathcal{D}_{\text{val}}$, network f_θ (ResNet-18 or VGG-16), epochs T=50, budget ϵ=4/255, step α=1/255, steps K=3
Output: final weights θ_T and best weights θ^*

for epoch $t = 1, \dots, T$ **do**
  for each mini-batch $(x, y) \subset \mathcal{D}$ of size 128 **do**
    $x \leftarrow$ random crop and horizontal flip of x
    switch Batch Normalisation to inference mode
    $x' \leftarrow \text{clip}_{[0,1]}(x + u)$, $u \sim \mathcal{U}(-\epsilon, \epsilon)^d$
    for $k = 1, \dots, K$ **do**
      $x' \leftarrow \Pi_{\mathcal{B}_\epsilon(x)\cap[0,1]^d}\big(x' + \alpha\,\text{sign}(\nabla_{x'} J(\theta, x', y))\big)$
    switch Batch Normalisation to training mode
    $L \leftarrow \frac{1}{2}J(\theta, x, y) + \frac{1}{2}J(\theta, x', y)$ ▷ Eq. (4)
    $\theta \leftarrow$ SGD step on L (momentum 0.9, wd 5×10^{-4})
  divide the learning rate by 10 if $t \in \{25, 37\}$
  $a_t \leftarrow$ PGD-10 (ϵ=8/255) accuracy on $\mathcal{D}_{\text{val}}$
  if a_t is the highest so far **then** $\theta^* \leftarrow \theta$
return θ_T (main result) and θ^* (secondary)

Algorithm 2 Evaluation and gap decomposition

Input: trained models f (2 networks $\times$ 2 objectives $\times$ 3 seeds), test set $\mathcal{D}_{\text{test}}$, budgets $\mathcal{E} = \{2, 4, 8\}/255$

for each model f **do**
  A_{clean}, Macro F1 on $\mathcal{D}_{\text{test}}$; $\mathcal{C} \leftarrow \{i : f(x_i) = y_i\}$
  for each $\epsilon \in \mathcal{E}$ **do**
    for attack $\in$ {FGSM, PGD-7, PGD-20} **do**
      $x_i^{\text{adv}} \leftarrow$ attack(f, x_i, y_i, ϵ); $A_{\text{rob}} \leftarrow$ accuracy
    $\rho \leftarrow |\{i \in \mathcal{C} : f(x_i^{\text{adv}}) = y_i\}|/|\mathcal{C}|$ (PGD-20)
    NRR by Eq. (5); logit-MSE ratio r over $\mathcal{C}$
  $A_{\text{AA}} \leftarrow$ AutoAttack on first 1,000 test images, ϵ=8/255
  check: PGD-20 at ϵ=64/255 gives 0%; $A_{\text{AA}} \le A_{\text{PGD-20}}$
average all metrics over seeds (mean $\pm$ std)
for each ϵ: split ΔA_{rob} by Eq. (7)

IV. Experimental Setup

Data and training. CIFAR-100 (50,000 training and 10,000 test images, 100 classes). We hold out 1,000 training images as a validation set. All models use SGD with momentum 0.9, batch size 128, weight decay 5×10^{-4} and initial learning rate 0.1, the optimiser settings of [12], for 50 epochs, with the learning rate divided by 10 after epochs 25 and 37. Data augmentation is random cropping with 4-pixel padding and horizontal flipping. Standard and adversarially trained models share all hyperparameters.

TABLE I

RESULTS ON CIFAR-100 AT $\epsilon = 8/255$ (ℓ_∞), FINAL CHECKPOINT, MEAN $\pm$ STD OVER SEEDS. ACCURACIES IN %, MACRO F1 IN $[0, 1]$.

Model	Clean Acc.	Clean F1	FGSM	PGD-7	PGD-20	AutoAttack
ResNet-18 (standard)	74.96±0.03	0.750	6.68±0.47	0.01±0.01	0.00±0.00	0.00±0.00
ResNet-18 (mixed AT)	69.81±0.04	0.697	17.37±0.12	10.27±0.30	7.10±0.16	5.77±0.31
VGG-16 (standard)	61.84±0.15	0.622	6.19±0.43	0.22±0.09	0.07±0.06	0.00±0.00
VGG-16 (mixed AT)	57.90±0.64	0.577	14.99±0.27	8.86±0.14	6.03±0.16	4.13±0.12

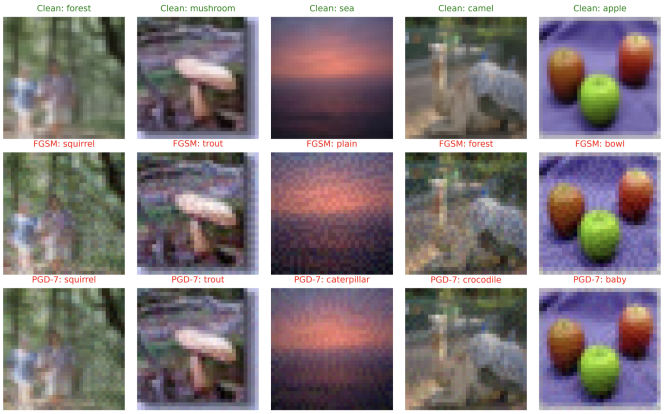


Fig. 2. Clean CIFAR-100 test images (top) and their FGSM (middle) and PGD-7 (bottom) versions at $\epsilon = 8/255$, with predictions of the standard ResNet-18 (green: correct, red: wrong). All five images are classified correctly before the attack and incorrectly after it.

Checkpoints. We report the final (last-epoch) checkpoint as our main result. As a secondary result we also report the best checkpoint, selected by clean validation accuracy for standard models and by PGD-10 ($\epsilon = 8/255$) validation accuracy for AT models, as in [12]. Section V-E explains why we do not use the best checkpoint as the main one.

Seeds and hardware. Each configuration is trained with 3 random seeds; we report mean $\pm$ standard deviation. Training ran on NVIDIA Tesla T4 GPUs in Google Colab and Kaggle (PyTorch 2.11 on Kaggle). Runs interrupted by session limits were resumed from the last saved epoch, restoring model, optimiser and scheduler state.

Sanity checks. Following standard practice [4], [18], we verify that PGD-20 with a very large budget ($\epsilon = 64/255$) drives accuracy of every model to 0%, and that AutoAttack is not weaker than PGD.

V. RESULTS

A. Overall Results and Evaluation Validity

Table I summarises all models at $\epsilon = 8/255$. The attacks are ordered as expected (FGSM $>$ PGD-7 $>$ PGD-20 $>$ AutoAttack), the large-budget sanity check reaches 0% for every model, and inside AutoAttack the black-box Square attack finds no additional adversarial examples beyond APGD (FAB-T adds a single example in one run). Together these checks indicate that the measured robustness is not caused by gradient masking. Fig. 2 shows typical adversarial examples. At $\epsilon = 8/255$ the perturbation is small but visible on close inspection at 32×32 resolution, which is why we also report the smaller budgets $2/255$ and $4/255$.

TABLE II

ACCURACY (%) UNDER PGD-20 AT DIFFERENT BUDGETS, FINAL CHECKPOINT. NRR FROM CLEAN AND PGD-20 ACCURACY (EQ. (5)); ρ: CONDITIONAL ROBUST ACCURACY (EQ. (6)).

Model	ϵ	PGD-20	NRR	ρ
ResNet-18 (std.)	2/255	3.06±0.20	5.9	4.1±0.3
VGG-16 (std.)	2/255	4.23±0.36	7.9	6.8±0.6
ResNet-18 (AT)	2/255	48.29±0.33	57.1	69.2±0.5
	4/255	28.17±0.37	40.1	40.3±0.5
	8/255	7.10±0.16	12.9	10.2±0.2
VGG-16 (AT)	2/255	37.05±0.63	45.2	63.9±0.4
	4/255	21.84±0.19	31.7	37.7±0.1
	8/255	6.03±0.16	10.9	10.4±0.4

TABLE III

DECOMPOSITION (EQ. (7)) OF THE PGD-20 GAP BETWEEN ADVERSARIALLY TRAINED RESNET-18 AND VGG-16 (POINTS), FINAL CHECKPOINT.

ϵ	Gap	Clean-acc. part	Conditional part
2/255	11.24	7.62 (68%)	3.62 (32%)
4/255	6.33	4.49 (71%)	1.84 (29%)
8/255	1.07	1.24 (116%)	−0.17

B. Standard Training

Standard training gives clean accuracies of 74.96% (ResNet-18) and 61.84% (VGG-16), but both models are almost defenceless. Already at $\epsilon = 2/255$, PGD-20 reduces accuracy to 3.06% and 4.23%, and at $\epsilon = 4/255$ to at most 0.5% (Table II, Fig. 3). At small budgets the standard VGG-16 is slightly more robust than the standard ResNet-18: 6.8% versus 4.1% of its correctly classified images survive PGD-20 at $\epsilon = 2/255$. Higher clean accuracy does not, by itself, imply higher robustness.

C. Effect of Mixed-Loss Adversarial Training

AT lowers clean accuracy by 5.15 points for ResNet-18 ($74.96 \to 69.81\%$) and by 3.94 points for VGG-16 ($61.84 \to 57.90\%$). In return, PGD-20 accuracy at the training budget $\epsilon = 4/255$ rises from 0.07% to 28.17% for ResNet-18 and from 0.50% to 21.84% for VGG-16, and at $\epsilon = 2/255$ to 48.29% and 37.05% (Table II). The protection transfers poorly to twice the training budget: at $\epsilon = 8/255$ PGD-20 accuracy is only 7.10% and 6.03% (AutoAttack: 5.77% and 4.13%), and only about 10% of correctly classified images survive.

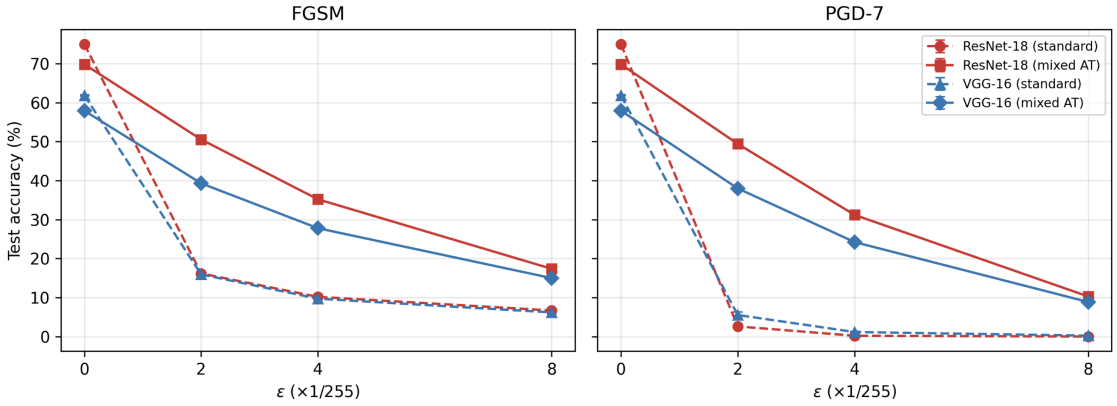


Fig. 3. Test accuracy of standard (dashed) and adversarially trained (solid) ResNet-18 and VGG-16 on CIFAR-100 versus perturbation budget ϵ, under FGSM (left) and PGD-7 (right). Final checkpoint, mean over 3 seeds; error bars (std) are smaller than the markers.

Robustness obtained with this objective is therefore largely specific to the budget used in training.

D. Architecture: Robustness or Clean Accuracy?

After AT, ResNet-18 has higher robust accuracy than VGG-16 at every budget (Table II). Table III splits this gap with Eq. (7). At $\epsilon = 2/255$ and $4/255$, about 70% of the gap comes from ResNet-18's higher clean accuracy and about 30% from a genuinely higher survival rate of correctly classified images ($\Delta\rho = 5.3$ and 2.6 points). At $\epsilon = 8/255$ the two networks have the same conditional robustness (10.2% vs. 10.4%), and the whole gap is explained by clean accuracy. Thus residual connections, or the other differences between the two networks, mainly make adversarial training reach a higher clean accuracy. They make the resulting model only moderately harder to attack at small budgets, and not at all at $\epsilon = 8/255$. Because the networks also differ in depth, parameter count and classifier head, we cannot attribute the remaining conditional advantage to skip connections specifically; controlled variants (e.g., VGG-16 with shortcuts) would be needed [13], [14].

E. Checkpoint Selection and Robust Overfitting

Table IV compares the best and final checkpoints of the AT models. The best checkpoint is slightly more robust (PGD-20 8.10% vs. 7.10% for ResNet-18, 6.46% vs. 6.03% for VGG-16; AutoAttack 0.7–0.8 points higher), a mild form of robust overfitting [20]. It is, however, much less stable for VGG-16: its clean accuracy varies by ± 2.54 points across seeds, compared with ± 0.64 for the final checkpoint, and it is on average 4.8 points lower. The cause is the selection signal: robust validation accuracy is about 7% on 1,000 images, so checkpoints that differ by one or two correctly classified images are ranked differently, and an early checkpoint with low clean accuracy can be selected. Measured with best checkpoints, the clean-accuracy cost of AT for VGG-16 would be 8.3 points instead of 3.9, overstating the trade-off by a factor of two. We therefore use the final checkpoint for the main comparison and recommend that studies selecting "best" checkpoints report the validation-set size and the variance this choice introduces.

TABLE IV
BEST VS. FINAL CHECKPOINT OF THE AT MODELS AT $\epsilon = 8/255$.

Model	Ckpt	Clean	PGD-20	AutoAttack
ResNet-18 (AT)	best	68.29±0.59	8.10±0.16	6.47±0.17
	final	69.81±0.04	7.10±0.16	5.77±0.31
VGG-16 (AT)	best	53.07±2.54	6.46±0.32	4.90±0.41
	final	57.90±0.64	6.03±0.16	4.13±0.12

TABLE V
PERCEPTION CONSISTENCY: RATIO $r = \mathrm{MSE}_{\mathrm{failed}}/\mathrm{MSE}_{\mathrm{defended}}$ OF LOGIT MSE BETWEEN CLEAN AND PGD-20 INPUTS, FINAL CHECKPOINT. $r < 1$ WOULD INDICATE THAT FAILURE CASES ARE LEARNED BETTER [12]. "–": FEWER THAN 30 DEFENDED SAMPLES.

Model	$\epsilon = 2/255$	$4/255$	$8/255$
ResNet-18 (std.)	2.74±0.16	–	–
VGG-16 (std.)	14.34±0.93	26.94±5.79	–
ResNet-18 (AT)	1.52±0.00	1.79±0.01	2.66±0.05
VGG-16 (AT)	4.58±0.17	5.92±0.24	8.22±0.73

F. Perception Consistency

Table V reports the ratio r of logit MSE between failed and defended samples. For standard models failed samples change their logits far more than defended ones ($r = 2.7$ for ResNet-18 and $r = 14.3$ for VGG-16 at $\epsilon = 2/255$), as expected. AT reduces the ratio for both networks, to 1.5 and 4.6 at $\epsilon = 2/255$, and ResNet-18 comes much closer to $r = 1$ at every budget. However, r stays above 1 in all our settings: the pattern reported by Wang and Liu [12] for PGD-AT, where failure cases have *smaller* logit change than defended ones, does not appear under our objective. A likely reason is that our AT is weaker (budget $4/255$, 3 PGD steps, half of the loss on clean data), so the over-sufficient learning of failure cases that they describe has not yet occurred. Our results are consistent with their view that AT increasingly equalises perception across samples, and suggest that the strength of this effect depends

on both the AT objective and the architecture. Absolute MSE values are not comparable across networks, because the logit scales differ; only ratios within one model are compared.

G. Limitations

(1) Our AT objective is weaker than the standard PGD-10 training at $\epsilon = 8/255$, which explains the low robustness at $8/255$; conclusions about $8/255$ should not be extrapolated to standard PGD-AT. (2) Training lasts 50 epochs, so clean accuracy is below the best reported values for these architectures, especially for VGG-16. (3) AutoAttack is run on 1,000 test images, giving a standard error of about 0.7 points at the measured accuracies. (4) We use one dataset and two architectures, and the architectures differ in more than skip connections. (5) Results are averaged over 3 seeds; small differences (< 1 point) should be read with caution.

VI. Conclusion

We compared ResNet-18 and VGG-16 on CIFAR-100 under an identical training recipe, with and without mixed-loss adversarial training, using FGSM, PGD and AutoAttack over several seeds. Without defence both networks fail almost completely at $\epsilon \geq 4/255$. Adversarial training at $\epsilon = 4/255$ gives substantial robustness at that budget (28.17% and 21.84% under PGD-20) for a clean-accuracy cost of 3.9–5.2 points, but little robustness at $\epsilon = 8/255$. ResNet-18 is more robust in absolute terms, yet about 70% of its advantage at small budgets, and all of it at $\epsilon = 8/255$, follows from its higher clean accuracy. We therefore suggest that architecture comparisons in adversarial robustness report conditional robust accuracy alongside absolute robust accuracy. We also showed that best-checkpoint selection on a small robust validation set can distort the measured trade-off, and that the perception pattern reported in [12] depends on the strength of adversarial training. Future work includes standard PGD-AT at $\epsilon = 8/255$, controlled architecture ablations, additional datasets, and testing whether trade-off methods such as RPAT [12] and CURE [11] change the decomposition reported here.

Acknowledgment

The first author is grateful to Professor Anwar Ghani of Nazarbayev University for his mentorship and guidance throughout this research.

References

[1] C. Szegedy, W. Zaremba, I. Sutskever, J. Bruna, D. Erhan, I. Goodfellow, and R. Fergus, "Intriguing properties of neural networks," in *Proc. ICLR*, 2014.

[2] I. J. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and harnessing adversarial examples," in *Proc. ICLR*, 2015.

[3] K. Eykholt, I. Evtimov, E. Fernandes, B. Li, A. Rahmati, C. Xiao, A. Prakash, T. Kohno, and D. Song, "Robust physical-world attacks on deep learning visual classification," in *Proc. IEEE/CVF CVPR*, 2018.

[4] N. Carlini and D. Wagner, "Towards evaluating the robustness of neural networks," in *Proc. IEEE Symp. Security and Privacy (SP)*, 2017, pp. 39–57.

[5] A. Athalye, N. Carlini, and D. Wagner, "Obfuscated gradients give a false sense of security: Circumventing defenses to adversarial examples," in *Proc. ICML*, 2018.

[6] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, "Towards deep learning models resistant to adversarial attacks," in *Proc. ICLR*, 2018.

[7] T. Bai, J. Luo, J. Zhao, B. Wen, and Q. Wang, "Recent advances in adversarial training for adversarial robustness," in *Proc. IJCAI*, 2021.

[8] D. Tsipras, S. Santurkar, L. Engstrom, A. Turner, and A. Madry, "Robustness may be at odds with accuracy," in *Proc. ICLR*, 2019.

[9] H. Zhang, Y. Yu, J. Jiao, E. P. Xing, L. El Ghaoui, and M. I. Jordan, "Theoretically principled trade-off between robustness and accuracy," in *Proc. ICML*, 2019.

[10] R. Rade and S.-M. Moosavi-Dezfooli, "Reducing excessive margin to achieve a better accuracy vs. robustness trade-off," in *Proc. ICLR*, 2022.

[11] S. Gowda, B. Zonooz, and E. Arani, "Conserve-update-revise to cure generalization and robustness trade-off in adversarial training," in *Proc. ICLR*, 2024.

[12] Y. Wang and L. Liu, "Failure cases are better learned but boundary says sorry: Facilitating smooth perception change for accuracy-robustness trade-off in adversarial training," in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2025. [Online]. Available: https://ieeexplore.ieee.org/document/11444296

[13] H. Huang, Y. Wang, S. M. Erfani, Q. Gu, J. Bailey, and X. Ma, "Exploring architectural ingredients of adversarially robust deep neural networks," in *Proc. NeurIPS*, 2021.

[14] S. Peng, W. Xu, C. Cornelius, M. Hull, K. Li, R. Duggal, M. Phute, J. Martin, and D. H. Chau, "Robust principles: Architectural design principles for adversarially robust CNNs," in *Proc. BMVC*, 2023.

[15] F. Croce, M. Andriushchenko, V. Sehwag, E. Debenedetti, N. Flammarion, M. Chiang, P. Mittal, and M. Hein, "RobustBench: A standardized adversarial robustness benchmark," in *Proc. NeurIPS Datasets and Benchmarks Track*, 2021.

[16] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE CVPR*, 2016, pp. 770–778.

[17] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *Proc. ICLR*, 2015.

[18] F. Croce and M. Hein, "Reliable evaluation of adversarial robustness with an ensemble of diverse parameter-free attacks," in *Proc. ICML*, 2020.

[19] E. Wong, L. Rice, and J. Z. Kolter, "Fast is better than free: Revisiting adversarial training," in *Proc. ICLR*, 2020.

[20] L. Rice, E. Wong, and J. Z. Kolter, "Overfitting in adversarially robust deep learning," in *Proc. ICML*, 2020.

[21] D. Wu, S.-T. Xia, and Y. Wang, "Adversarial weight perturbation helps robust generalization," in *Proc. NeurIPS*, 2020.

[22] Y. Wang, L. Li, J. Yang, Z. Lin, and Y. Wang, "Balance, imbalance, and rebalance: Understanding robust overfitting from a minimax game perspective," in *Proc. NeurIPS*, 2023.

[23] Y.-Y. Wu, H.-J. Wang, and S.-T. Chen, "Annealing self-distillation rectification improves adversarial training," in *Proc. ICLR*, 2024.

[24] Y. Dong, K. Xu, X. Yang, T. Pang, Z. Deng, H. Su, and J. Zhu, "Exploring memorization in adversarial training," in *Proc. ICLR*, 2022.

[25] A. Abuduweili and C. Liu, "Estimating neural network robustness via Lipschitz constant and architecture sensitivity," arXiv preprint arXiv:2410.23382, 2024.